

JomExam

ESAT Mathematics 2 — Sample Paper

Six questions in the style of ESAT Mathematics 2. Allow 9 minutes — the real paper's 90-second pace. No calculator. One option is correct in each question, and there is no negative marking.

Worked solutions with the trap named follow the questions. The full experience — 27 questions, 40 minutes, and a report naming the skill behind every wrong answer — is free at jomexam.com.

Question 1. The tangent to the curve $y = x^3 - 4x + 1$ at the point where $x = 2$ is drawn. What is the y-intercept of this tangent?

- A 1
- B -15
- C 8
- D -7
- E 15

Question 2. Evaluate the area under the curve $y = 6x^2$ between $x = 1$ and $x = 2$.

- A 14
- B 12
- C 21
- D 7
- E 42

Question 3. Given that $2 \log_{10} x = \log_{10} 9 + \log_{10} 4$, and $x > 0$, what is the value of x ?

- A 36
- B 18
- C 13
- D 6
- E ± 6

Question 4. How many solutions does the equation $\sin^2 \theta = 3 \cos^2 \theta$ have for $0 \leq \theta \leq 2\pi$?

- A 1
- B 2
- C 4
- D 3
- E 8

Question 5. The sum of the first n terms of a series is given by $S_n = n^2 + 3n$. What is the eighth term of the series?

- A 88
- B 18
- C 70
- D 11
- E 24

Question 6. The curve $y = f(x)$ has a minimum point at $(2, -5)$. What are the coordinates of the minimum point of the curve $y = 3f(x - 1)$?

A $(1, -15)$

B $(3, -5)$

C $(3, -15)$

D $(1, -5)$

E $(6, -15)$

Worked solutions

Question 1 — answer B.

1. $dy/dx = 3x^2 - 4$; at $x = 2$ the gradient is 8.
2. The point: $y(2) = 8 - 8 + 1 = 1$, so the tangent is $y - 1 = 8(x - 2)$.
3. $y = 8x - 15$: the y-intercept is -15 .

The trap: The trap is 1 — the curve's own value at $x = 2$ offered as the intercept. The tangent's intercept lives at $x = 0$, and the question is only answered when the line is actually built.

Question 2 — answer A.

1. $\int 6x^2 dx = 2x^3$.
2. Evaluate: $2(8) - 2(1) = 16 - 2 = 14$.

The trap: The trap is 12 — differentiating instead of integrating and evaluating $12x$ at $x = 1$. Under time pressure the two operations swap silently; the power should RISE for an area.

Question 3 — answer D.

1. $2 \log x = \log x^2$, and the right side is $\log 36$.
2. $x^2 = 36$, and since $x > 0$: $x = 6$.

The trap: The trap is 18: adding logs by adding their arguments. Logs add when arguments MULTIPLY — and the $x > 0$ condition quietly removes the $\pm$ that option E dangles.

Question 4 — answer C.

1. Divide by $\cos^2 \theta$ (checking $\cos \theta = 0$ gives $\sin^2 \theta = 3$, impossible — so nothing is lost): $\tan^2 \theta = 3$.
2. $\tan \theta = \pm\sqrt{3}$: solutions at $\pi/3, 2\pi/3, 4\pi/3, 5\pi/3$.

The trap: Two exits, both guarded: taking only $\tan \theta = +\sqrt{3}$ gives 2, and the division by $\cos^2 \theta$ is safe here — but only because you checked. The check is one line; skipping it is a habit that costs marks elsewhere.

Question 5 — answer B.

1. The eighth term is $S_8 - S_7$.
2. $S_8 = 64 + 24 = 88$; $S_7 = 49 + 21 = 70$; the term is 18.

The trap: The trap is 88 — the SUM of eight terms reported as the eighth term. S_n is a running total; individual terms are differences of it.

Question 6 — answer C.

1. $f(x - 1)$ shifts the curve 1 to the RIGHT: the minimum moves to $x = 3$.
2. Multiplying by 3 stretches vertically: -5 becomes -15 .
3. Minimum: $(3, -15)$.

The trap: Both traps are one coordinate each: $(1, -15)$ shifts left instead of right — the minus inside the bracket moves the curve against instinct — and $(3, -5)$ forgets that a vertical stretch scales a NEGATIVE minimum further down.